5.1. Human Robot Interaction Challenge

Description

The robot is working as a receptionist at a party. The robot has to take two new guests to the living room, offer a free place to sit and introduce them to eachother. The second guest brings a bag for the robot to carry.

Main goal: The robot welcomes and assists two newcomers at a party and offering them a seat, and maintaining appropriate gaze direction during conversation (at person speaking, direction of navigation). The second guest brings a bag to be placed somewhere in the house.

Optional goals:

1. Detect doorbell sound as a signal that a guest has arrived.
2. Open the entrance door for each arriving guest.
3. Visually describe the first guest to the second guest before reaching the living room.
4. Understand each guest's name and favorite drink without asking any confirmation or correction questions (non-essential questions).

Focus

System Integration, *Human-Robot Interaction*, *Person Detection*, *Person Recognition*

Setup

- **Location**:
  - The test takes place in the living room.
  - The robot starts inside the *Arena* at a predefined location near where guests are sitting.
  - **Entrance:** The entrance door is open by default. The team leader can request it to be closed to allow the robot to earn additional points by opening it for the guests.
- **People:**
  - **Arriving Guests:** Both guests have a name and a favorite drink. An arriving guest will wait by the door, and the robot has to move to meet guests by the door. A guest will not move inside the house until the robot has initiated conversation. It is not allowed to instruct guests to move in front of the robot to recognize the guest and start the conversation. The robot is the one that must approach the guest. Guests have to be guided to the living room where the robot will offer a free place to sit and introduce the guests to each other. Each guest will arrive separately.
  - **Passive Guests:** A few individuals, including possibly spectators, can be anywhere in the arena — sitting or standing in relevant places, e.g., on the couch, near the seats, or in the area with the drinks.

Procedure

Both guests arrive separately. A doorbell will ring to signal the guest has arrived. The doorbell will only ring if the robot is waiting for a new guest at the starting position. The robot either

opens the door for the guest or meets them at the door. It greets the guest and asks for their name and favorite drink. The robot escorts the guests to the living room and offers a free seat. Once both guests are seated, the robot introduces them to each other. To introduce the two guests, the robot must (for both guests) look at a guest and state the other arriving guest name and favorite drink. The second guest will bring a bag for the host. The robot should ask for the bag to be handed over and move to the host. The host is already waiting and instructs the robot to follow him/her. The host will move to a random place in the house and inform the robot it can drop the bag there.

- **Doorbell:** A doorbell sound will be played when a guest is by the door ready to be met. The doorbell sound will not be provided beforehand and may be any traditional doorbell sound. As an alternative, if no doorbell sound is available, the guest can knock on the door. Waiting for the doorbell is optional, robots can bypass it and directly meet the guest at the door. In order for the doorbell sound to be considered detected, the robot must be waiting at the starting position and react instantly to the sound.
- **Greeting guests:** The names of the guests can be any popular english names. The favorite drinks of the guests can be any english named popular drink (it may not be in the list of objects provided). The guest paces slightly to the left and right during the conversation with the robot. Other people might appear in the background.
- **Looking at person:** The robot must look at the person it is talking to. Guests will move slightly to confirm the robot dynamically follows them with its gaze. Points for gaze tracking are awarded only if the robot continuously tracks the moving person.
- **Looking at direction of navigation:** The robot must look in the direction it is navigating. Persistently looking in an unrelated direction or at unrelated people during navigation results in a score deduction.
- **Seating People:** The robot must clearly indicate or point to a location or seat for each guest.
- **Switching Places:** After being seated, guests may switch seats.
- **Introductions:** When introducing guests, the robot must clearly identify the person being introduced and state their name and favorite drink. The robot must be looking at a guest and state the other arriving guest name and favorite drink. The robot must do this for both guests. Introducing two people means to introduce them to each other.
- **Non-Essential Questions:** If the robot does not ask any non-essential questions to help understand, confirm, or correct information, bonus points are awarded. Questions such as "Did you say your name is James?" or "Is your favorite drink milk?" are considered non-essential questions. However, asking a guest to repeat an answer because the robot could not hear is not considered a non-essential question.
- **Bag handover:** The robot must grab the bag from the second guest hand. If this is not possible, the robot can instruct the guest to place the bag somewhere in its structure
- **Follow the host:** The host will be sitting in the living room. The robot must inform the host that it has a bag and ask the host for guidance. When the robot is ready to follow, it should inform the host. The host then walks naturally toward the designated bag drop location. Once at the correct location, the host informs the robot that it should place the bag on the floor.

Additional rules and remarks

1. **Opening Door Timing:** The Time of the test only starts after the first person enters the arena or 1 minute after start signal.
2. **Misunderstanding:** Not understanding the guests and asking them again is fine. Continuing with a wrong name or favorite drink causes a score reduction.
3. **Partial Scoring:** The main task allows partial scoring per guest.
4. **Additional Seating:** Referees may add chairs to the living room area.
5. **Human Assistance:** Score reduction applies per guest as follows:
   - **Custom Operator:** Since the main focus of the test is HRI, no custom operator can be chosen.
   - **Alternative HRI:** Using an alternative HRI to understand a guest causes a score reduction per information (name or favorite drink).
   - **Recognizing People:** If the robot has to ask for help to identify people, the score is reduced.
   - **Follow the host:** Penalties apply if the robot loses the operator and needs help recovering. The following forms of assistance are penalized:
     5.1. Natural interaction (e.g., waving and calling)
     5.2. Unnatural interaction (e.g., raising both hands and jumping)
     5.3. Returning in front of the robot to resume following.
     5.4. Physical contact (e.g., pulling the robot's hand)
6. **Bag handover:** If the robot has to ask the guest for help during handover, the score is reduced. If the robot tells the guest to place the bag somewhere on its structure, an additional penalty is applied. To score the maximum amout of points:
   - The robot must clearly signal its readiness to receive the item (e.g., by holding its manipulator still, using lights, audio cues, or verbal prompts).
   - The manipulator must be positioned atleast within 10 cm of the operator's hand.
   - The operator can complete the handover; the robot should not reach, grasp, or initiate dangerous movement toward the human.
   - Explicit verbal guidance from the operator (e.g., *"Place the item in my gripper, I'll close it in 3. . . 2. . . 1"*) will result in penalties.
   - The robot may, however, issue saefty cues such as: *"I detect the object in my hand. I will close my gripper in 3 seconds, please be careful."*.

Instructions

To Referee

The referees need to:

- Assign name and favorite drink to two volunteers.
- Arrange and rearrange people in the living room.
- Open the entrance door upon request from the robot.

To OC

During setup day:

- Announce the bag that will be used for handover.

At least two hours before test:

- Announce the starting position.
- Recruit four volunteers: two as arriving guests and two as passive guests.

Score sheet

The maximum time for this test is **6:00** minutes.

Action	**Score**
Main Goal	
Offer a free seat to the new guest	*2×100*
Look in the direction of navigation or at the navigation goal	*2×15*
Look at the person talking, when receiving a guest	*2×50*
Introduce both guests to each other: correctly say name and favorite drink of each guest	*4×30*
While introducing guests, look to the correct guest while talking about the other guest	*2×50*
Grab the bag via handover from the guest	*50*
Following the host to the bag drop area	*200*
Drop the bag on the correct area	*50*
Bonus Rewards	
Detect the doorbell sound as a signal that a guest has arrived	*2×30*
Open the entrance door for a guest	*2×200*
Describe the first guest to the second guest (per correct visual attribute)	*4×20*
Not asking non essential questions to confirm or correct information	*4×15*
Penalties	
Wrong guest information was memorized (continue with wrong name, drink or interest)	*4×–40*
Describe the first guest to the second guest (per incorrect visual attribute)	*4×–20*
Alternative HRI	*6×–20*
Not recognizing people	*2×–200*
Penalties - Guiding	
Ask the guest to place the bag somewhere on the robot	*–25*
Drop the bag while following the host	*–50*
Ask the host to place the bag on the bag drop location	*–50*
Rediscovering the operator by natural interaction	*–50*
Asking operator to wait	*–50*
Guiding the robot with physical contact (take by the hand)	*–150*
Special Penalties & Bonuses	
Not attending (see sec. 3.8.1)	*–500*
Outstanding performance (see sec. 3.8.3)	*145*

Total Score (excluding special penalties & standard bonuses) *1450*

5.2. Pick and Place Challenge

This challenge evaluates manipulation capabilities through cleaning and organizing the kitchen and preparing a simple breakfast.

Main goals:

1. Tidy up all objects on the dining table:
   1.1. Place dirty tableware and cutlery inside the dishwasher.
   1.2. Place designated trash items in the trash bin.
   1.3. Store other objects in the cabinet, grouping them with similar items.
2. Set up breakfast on a clean area of the table, including a bowl, spoon, cereal, and milk.

Optional goals:

1. Pick up trash from the floor.
2. Open and close the dishwasher door.
3. Pull and push the dishwasher rack.
4. Place a dishwasher tablet inside the dishwasher.
5. Pour milk and cereal into the breakfast bowl.

Focus

Object perception, *manipulation in narrow spaces*, and *task planning*.

Setup

- **Locations:**
  - **Start Location:** The robot waits outside the arena and enters when the door is opened.
  - **Test location:** The test takes place in the kitchen.
- **People:**
  - No people are involved unless the robot explicitly requests assistance.
- **Furniture:**
  - **Dishwasher:** Located near the dining table. Closed by default; the robot may request help to open or close doors or racks.
  - **Trash bin:** A trash bin is located in the kitchen.
  - **Cabinet:** The cabinet contains objects arranged by category or similarity on different shelves.
- **Objects:**
  - **Table setting:** Seven objects arranged on the table in a typical post-meal setting, possibly stacked:
    * *Cutlery*: One piece (fork, knife, or spoon).
    * *Tableware*: One mug or cup and one plate.
    * *Trash*: One trash item. One object category will be treated as trash for this task.
    * *Other objects*: Two items not belonging to the above categories.

- **Breakfast items:** The breakfast items are a bowl, a spoon, milk and cereal. The object distribution is as follows:
  - *Bowl and Spoon*: On top a designated surface in the kitchen.
  - *Milk and Cereal*: Inside the cabinet, next to their respective categories.
- **Cabinet objects:** A doorless cabinet. Each side of the shelves contains objects arranged in groups, either by category or likeliness.
- **Dishwasher tab:** The tab can be found on top of a designated surface and should be placed inside the dishwasher slot.
- **Floor Object:** One trash item will be placed near the trash bin.

Procedure

1. **Test start:** The robot moves to the kitchen when the arena door is open.
2. **Table clean up:** The robot tidies up the table by putting: the cutlery and tableware items inside the dishwasher, the trash in the trash bin and the other objects in the cabinet, grouping them by category or similarity.
3. **Serve breakfast:** The robot sets the table for breakfast by placing the bowl, spoon, cereal and milk on the table in a typical setting for a meal. There needs to be a comfortable amount of free space around the breakfast.
4. **Sequence:** The robot is free to determine the order and method for performing the pick-and-place tasks. There is no predefined sequence, the robot may execute them in any way it finds optimal.

Additional Rules and Remarks

1. **First Pick Bonus:** To encourage manipulation, the robot receives an additional bonus for successfully picking the first object during the test. This bonus is awarded only once.
2. **Designated Location:** All objects designated location correspond to the appropriate furniture, either on top of or inside.
3. **Safe placing:** Objects must be placed with care, namely the robot should place rather than throw or drop objects.
4. **Dishwasher door:** The dishwasher door is closed by default. The robot may ask for help to open or close the door or racks at any time during the task. If the robot fails to open/close the door/rack, it must clearly state this and request the referee to open/close the door/rack.
5. **Correct dishwasher item placement:** Items must be correctly positioned in the rack, as a human would place them.
6. **Incorrect cabinet category categorization:** Objects must be grouped with similar items. Misplaced items incur score reductions. Objects that do not semantically belong to any of the categories represented on the shelves should put in an empty part of the shelf.
7. **Breakfast placement:** The table must be set in a typical setting for a meal. The spoon must be placed next to the bowl, and the cereal and milk must be placed next to each other.
8. **Breakfast area cleanliness:** The area immediately surrounding the breakfast items on the table must be kept clear of any other objects. Items too close at (5cm) or cluttering the space, will result in a score penalty for the breakfast placement.

9. **Pouring:** A significant amount of the milk and cereal must be poured, Pouring a couple of drops of milk or bits of cereal is not enough.
10. **Human Assistance:** Scores are reduced if the robot receives help, such as pointing to objects, handing objects to the robot, or repositioning items. Assistance with opening the milk container or moving parts of the dishwasher does not incur a penalty.
11. **Communicating Perception**: The robot must clearly indicate its perception to the referee. Pointing, attempting to pick objects, or visualizing one object at a time is sufficient. If visualization is utilized, the surrounding scene must remain visible and the robot needs to announce and confirm the referee perceived the visualization.
12. **Trash:** One object category will be treated as trash for this task. Announced during *Setup Days*.

OC Instructions

During the *Setup Days*:

- Provide official cutlery and tableware.
- Provide official objects.
- Designate a trash category.
- Announce the table and cabinet used for the test.
- Announce locations of dishwasher tablet, bowl, and spoon. (in the kitchen)

Referee Instructions

The referee needs to:

- Place table objects (1 cutlery, 1 plate, 1 mug or bowl, 1 trash, 2 other objects).
- Place one trash on the floor (near trash bin).
- Arrange cabinet objects by category or similarity.
- Place the bowl and spoon and dishwasher tab.
- Place the milk and cereal inside the cabinet next to their respective category.

Scoresheet

The maximum time for this test is **7:00** minutes.

Action	**Score**
Navigate to the table	*15*
Correctly recognize an object	*12×10*
Perceive objects on a shelf and indicate the correct placement	*2×30*
Picking	
Picking up an object for transportation	*12×50*
From the floor	*+30*
Cutlery	*2×+50*
Plate	*+100*
Dishwasher tablet	*+100*
Placing	
Place an object in its designated location	*12×40*
Place an item correctly in the dishwasher (cleanable and in a human-like manner)	*3×+70*
Place an object next to similar objects in the cabinet	*2×+20*
Bonus Rewards	
First Pick Bonus	*100*
Pull or push the dishwasher rack	*2×100*
Open or close the dishwasher door without assistance	*2×200*
Place the dishwasher tablet in its designated slot inside the dishwasher	*200*
Open milk container without assistance	*400*
Pour cereal or milk into the bowl without assistance	*2×200*
Penalties	
Objects thrown or dropped while placing	*12×–40*
Area around breakfast items is not cleaned (from breakfast placement scores)	*4×–30*
Breakfast not served in a typical meal setting	*–50*
Objects dropped on the floor	*12×–40*
Spilling cereal and milk while pouring	*2×–100*
Human assistance: object repositioned by a person	*12×–40*
Human assistance: handover	*24×–100*
Human assistance: opening milk container	*-0*
Human assistance: moving dishwasher door or rack	*-0*
Special Penalties & Bonuses	
Not attending (see sec. 3.8.1)	*–500*
Using alternative start signal (see sec. 3.6.8)	*–100*
Outstanding performance (see sec. 3.8.3)	*355*

Total Score (excluding special penalties & standard bonuses) *3555*

5.3. General Purpose Service Robot

The robot is asked to understand and execute commands requiring a wide range of different abilities.

Main Goal: Execute three commands issued by the operator.

Focus

Task planning, object/people detection and recognition, object feature recognition, object manipulation

Setup

- **Locations:**
  - **Task location:** The task takes place inside the *Arena*. Commands may require the robot to leave the room. The *Arena* is in its nominal configuration for this task.
  - **Start location:** The robot starts outside the *Arena*. When the door opens, it must navigate towards the *Instruction Point*.
  - **Instruction point:** The robot returns to this point after completing all the commands.
- **People:**
  - **Professional Operator:** The referee issues standard commands to the robot. If the robot consistently fails to understand the command (e.g. after three tries), teams can use a custom operator.

Procedure

1. **Instruction point:** At least two hours before the test, the referees announce the location of the *Instruction Point*.
2. **Command execution:** The robot will decide how the commands will be issued and advise the operator, i.e., either consecutively or one-by-one. If the commands are issued one-by-one, the robot must return to the operator after completing each task.
3. **Back to the instruction point:** The robot goes back to the *Instruction Point* after completing all the commands given by the operator.
4. **Pausing the Timer:** The referee pauses the timer as soon as the robot reaches the instruction point to reset the arena for the next command. The timer resumes once the referee signals the start of the next command.

Additional Rules and Remarks

1. **Interleaved Task Bonus:** The robot receives an additional bonus if it successfully completes commands in an interleaved order rather than strictly consecutively. This bonus is awarded only when all three commands are received at once. The interleaved execution

must be meaningful, for example by saving time or reducing unnecessary movements. *Example:* The robot first picks up an object, then performs another task along the way, and only afterward delivers the object to its original destination.

2. **Partial Scoring:** The solution allows partial scoring.
3. **Command generator:** Tasks will be generated using the official command generator[1]. Once a command has been generated it will be entered into an LLM to re-generate a similar phrase, e.g. the generated command is "get me a coke from the kitchen" re-phrased command is "Go to the kitchen, find a coke, and bring it to me". Each command may be re-phrased up to 3 times getting simpler with each rephrasing.
4. **Test start:** The robot moves to the *Instruction Point* when the arena door is open.
5. **Team Coaching:** Teams are not allowed to coach, or instruct the operators. Doing so results in disqualification from the task.
6. **Custom Operators:** If a custom operator is used they can only choose between the three re-phrased commands to give.
7. **Autonomy Skip:** Score reductions apply in the following cases:
   - Use of a custom operator.
   - Bypassing speech recognition.
   - Receiving human assistance to accomplish a command.
   - Instructing a human assistant to perform the whole task.
   - QR codes will not be available.

Referee Instructions

- Provide the commands to the operators.

OC Instructions

At least two hours before the test:

- Generate the robot commands and pass through LLM to get a similar command (do not reveal them to the teams).
- Announce the location of the instruction point.
- Recruit volunteers to assist during the test.

During the test:

- The arena will be setup for all command executions.

Score Sheet

The maximum time for this test is **7:00** minutes.

[1] https://github.com/RoboCupAtHome/CommandGenerator

Action	Score
Main Goal	
Understand the spoken command	*3×80*
Demonstrate a plan has been generated	*3×100*
Solving the command	*3×250*
Bonus Rewards	
Interleaved Task Bonus	*200*
Penalties	
Using a custom operator	*3×–20*
Request a rephrasing	*6×–30*
Bypassing speech recognition	*3×–50*
Human assistance: will apply a percentage penalty according to similar penalties in other tests.	*3×–250*
Special Penalties & Bonuses	
Not attending (see sec. 3.8.1)	*–500*
Using alternative start signal (see sec. 3.6.8)	*–100*
Outstanding performance (see sec. 3.8.3)	*149*

Total Score (excluding special penalties & standard bonuses) *1490*

5.4. Doing Laundry

The robot assists with laundry by retrieving clothes from a washing machine and folding them neatly on a table. This test evaluates deformable object manipulation, appliance interaction, and task sequencing.

Main goal: Transport clothes to a table, and fold them.

Optional goals:
1. Opening the washing machine door
2. Retrieving clothes from inside the washing machine
3. Using a laundry basket for transportation
4. Folding multiple pieces of clothing and stacking them

Focus

Deformable object manipulation, *bimanual grasping.*

Setup

- **Locations:**
  - **Start location:** Before the test, the robot waits outside the *Arena* and navigates to the laundry area when the door is open.
  - **Test locations:** The testing area contains a washing machine, a folding surface nearby, and a laundry basket.
- **People:**
  - No people are involved in the test, unless the robot requires human assistance.
- **Furniture:**
  - **Laundry basket:** A lightweight basket requiring two arms to carry, placed near the washing machine. The basket already has 6–8 pieces of clothing placed inside.
  - **Washing machine:** Positioned with its door closed at the start. Clothes are placed inside. There are 2–4 pieces of clothing in the washing machine.
  - **Table:** Positioned near the washing machine. Used as the folding surface.
- **Objects**:
  - **Clothes:** A set of 8–12 pieces of laundry made from cloth or other deformable fabric.

Procedure

1. **Location announcement:** At least two hours before the test, referees announce which washing machine and table will be used, along with the approximate location of each.
2. **Test start:** The robot enters the testing area once the arena door is opened.
3. **Retrieving laundry:** The robot retrieves the clothes from the laundry basket or directly from the washing machine and places them on the table.
4. **Folding:** The robot attempts to fold at least one piece of clothing neatly.

Additional rules and remarks

1. **Laundry Types:** The laundry will consist exclusively of T-shirts.
2. **Cloth placement:** Clothes must be placed on the table before folding. Folding on the floor is not allowed.
3. **Folding quality:** Folding is evaluated by neatness and whether the cloth is flattened and stacked.
4. **Multiple items:** Additional points are awarded for folding multiple pieces of clothing and stacking them.
5. **Human Assistance:** Scores are reduced if human assistance is received, in particular for:

    - opening the washing machine door
    - handing clothes to the robot
    - **Folding:** Penalties are proportional to the amount of help provided:
        - Minimal help (e.g., smoothing wrinkles, stabilizing cloth): small penalty (100 points).
        - Partial folding (e.g., folding one half or aligning edges): moderate penalty (200 points).
        - Major help (e.g., completing entire fold/stack): maximum penalty.

OC Instructions

At least two hours before the test:

- Announce which washing machine and table will be used.

Referee Instructions

The referee needs to:

- Ensure the washing machine and basket contains the laundry before the test begins.
- Close the washing machine door at the start.
- Verify that the folding table is clear before the test.

Score sheet

The maximum time for this test is **7:00** minutes.

Action	Score
Main Goal	
Navigating to the laundry area	*15*
Picking up a clothing item	*100*
Placing clothing on the table	*100*
Folding a piece of clothing	*400*
Bonus Rewards	
Opening the washing machine door	*300*
Retrieving clothes from inside the washing machine (per item)	*4×100*
Using the basket for transportation	*300*
Folding additional clothes (per item)	*6×300*
Stacking folded clothes neatly (per item)	*6×200*
Penalties	
Human Assistance: opening the washing machine door	*–300*
Human Assistance: handing clothes to the robot	*–100*
Human assistance during folding will apply a percentage penalty according to the amount of help	*–400*
Special Penalties & Bonuses	
Not attending (see sec. 3.8.1)	*–500*
Using alternative start signal (see sec. 3.6.8)	*–100*
Outstanding performance (see sec. 3.8.3)	*461*

Total Score (excluding special penalties & standard bonuses) *4615*

5.5. Restaurant

Description

The robot is required to retrieve and serve orders to customers in a real, previously unknown restaurant environment.

Main goal: Detect calling or waving customer, autonomously navigate to their tables, take their orders, and serve them accordingly.

Optional goal: Use an unattached tray to transport the order.

Focus

This task focuses on *Task planning*, *Online mapping*, *Navigation in unknown environments*, *Gesture detection*, *Verbal interaction* and *Object manipulation*.

Setup

- **Locations:**
  - This task takes place in a real restaurant fully equipped and in business. If this is not possible, an alternative venue may be selected, provided it resembles a restaurant environment and is outside the *Arena*.
  - The *Restaurant* location will remain undisclosed until the start of the test.
  - The robot starts next to the *Kitchen-bar* (a designated table located near the restaurant's kitchen), facing the customer seating area (dining tables).
- **People:**
  - A *Professional Barman* (member of the *technical committee* (TC)) awaits at the other side of the *Kitchen-bar* for orders to be placed. The *Professional Barman* assists the robot on request.
  - There may be real customers and staff present.
  - There are at least three tables occupied by professional customers (member of the *organizing committee* (OC), *technical committee* (TC) or other teams).
  - There are at least two tables occupied with regular customers.
  - Customers may call the robot any time, even simultaneously.
- **Furniture:** The furniture is not standardized and remains unchanged during the test.
- **Objects:**
  - Objects to fulfill orders are located on the *Kitchen-bar*.
  - Each order consists of 2–3 objects, randomly selected.
  - All edible/drinkable objects from the list of standard objects (see Section 3.2.5) are eligible to be part of the orders.

Procedure

1. The referee instructs the team to bring the robot to the start location.
2. The referee gives the start signal and starts the timer.
3. The team leaves the area after the start signal.

4. A *technical committee* (TC) member follows the robot ready to press the emergency stop button.
5. The robot detects a calling or waving customer and navigates to their table.
6. The robot take the customer's order, place the order, and deliver it.
7. **Optionally**, the robot may use an unattached tray to transport the order.

Additional rules and remarks

- **Remarks:**
  - This test occurs in a public area. Any physical contact with people or furniture results in immediate emergency stop.
  - Since this task is performed outside the arena, the time limit may be longer than the others tasks.
  - The availability of wireless, external computing devices, or electrical outlets can't be guaranteed. Assume unavailability.
  - The robot must interact with the customers directly. Teams are not allowed to assist or instruct them.
  - The robot may use up to two minutes to instruct the *Professional Barman* per order.
  - Examples of instructions include, but are not limited to:
    * Request guidance to a customer's table.
    * Ask for pointing direction.
    * Ask to press on the screen.
  - The robot may:
    * Take and place several orders before delivery,
    * Alternate between order-taking and serving, or
    * Handle one order at a time.
  - The robot should politely confirm the order to the client when receiving it, keeping the guest pleased.
  - The robot can either transport each object individually, or using a tray. All delivered objects must be placed on the customer's table.
  - For tray delivery, the robot must:
    * Place the items on the tray,
    * Pick up and transport the tray,
    * Deliver the items by either directly placing them on the table or placing the tray first.
  - If requested, the *Professional Barman* will place the order in a basket or tray for the robot to deliver it.
  - Only two team members are allowed near the robot upon arrival for watching and charging.
  - If audience interference makes task execution impossible, the team may immediately repeat the test.
  - Each Deus Ex Machina penalty for skipping manipulation is capped at twice per order so receiving an order with three objects is not more punishing.
  - If the robot detects a customer but does not reach their table, it must clearly identify the person (e.g., show a picture) to earn partial points.

  - When at the front of the queue, teams may begin startup procedures with the robot stationary. Once called, the robot must be brought directly and steadily to the start location—only minor adjustments are allowed (no back-and-forth or full turns).

- **Disqualification:**
  - Touching the robot after the start signal.
  - Mapping the environment in advance.

Instructions:

To Referee

The referee needs to:

- Prepare orders for each client.

To OC

The OC needs to:

- **During Setup days:** Check with local (security) management if the possible location, including a sufficient queuing area, can be used for the restaurant test.
- **1 hour before the test:** Gather all teams and robots to move to some nearby queuing area and instruct the teams how/when to move to the actual test location.

Score sheet

The maximum time for this test is **15:00** minutes.

Action	**Score**
Regular Rewards	
Detect calling or waving customer	*2×100*
Reach a customer's table without prior guidance/training	*2×100*
Understand and confirm the order received to the customer	*2×200*
Communicate the order to the barman	*2×100*
Picking up the requested item from the *Kitchen-bar*	*2×200*
Return to the customer table with the order	*2×100*
Serve the order to the customer	*2×200*
Bonus Rewards	
Use an unattached tray to transport	*2×200*
Regular Penalties	
Being guided to a table	*2×−100*
Not making eye-contact when taking an order	*2×−80*
Not reaching the bar (barman has to move from behind the bar to interact with the robot)	*2×−80*
Asking for directional confirmation	*2×−80*
Deus ex Machina Penalties	
Asking the Barman to handover object to the robot	*4×−100*
Guest needing to take the object from a tray or the robot's hand	*4×−100*
Being told/pointed where is a table/*Kitchen-bar*	*2×−100*
Special Penalties & Bonuses	
Not attending (see sec. 3.8.1)	*−500*
Using alternative start signal (see sec. 3.6.8)	*−100*
Outstanding performance (see sec. 3.8.3)	*240*

Total Score (excluding special penalties & standard bonuses) *2400*